1 M1 for $3(-4)^2$ or $-5(-4)(2)$ or 48 or $+40$

88 **[A1]**

[Total Marks: 2]

2 Correct substitution: $5(-2)^2 - 3(4)$ or $5(4) - 12$ **[M1]**

8 **[A1]**

[Total Marks: 2]

3 $3(4) - 2(-5)$ or $12 - -10$ or $12 + 10$ **[M1]**

22 **[A1]**

[Total Marks: 2]

4 Substitutes values correctly, e.g. $3(-3)^2 - 4(-5)$ **[M1]**

47 **[A1]**

[Total Marks: 2]

5 Substitution of all three values correctly e.g. $\dfrac{3(5) - (-4)^2}{2(0.5)}$ **[M1]**

-1 **[A1]**

[Total Marks: 2]

6 Substitute the given values into the formula: $v = 5 + (-2)(3)$ **[M1]**

$v = 5 - 6$ **[M1]**

-1 **[A1]**

[Total Marks: 3]

7 (a) $4(-2) - 3(5)$ or $-8 - 15$ **[M1]**

-23 **[A1]**

(b) $\dfrac{5^2 - 7}{2(-3)}$ or $\dfrac{25 - 7}{-6}$ **[M1]**

-3 **[A1]**

[Total Marks: 4]

8 (a) -6 or -20 seen **[M1]**

-26 **[A1]**

(b) $(-3)^2 = 9$ or $4(-3) = -12$ seen **[M1]**

-3 **[A1]**

[Total Marks: 4]

9 (a) $P = 3(4)^2 - 2(4)(-3) + 5(-3)$ or $48 + 24 - 15$ seen **[M1]**

57 **[A1]**

(b) $19 = 3(-2)^2 - 2(-2)b + 5b$ or $19 = 12 + 9b$ **[M1]**

$\frac{7}{9}$ **[A1]**

[Total Marks: 4]

10 (a) Substitutes values correctly: $5(3) - 2(-4)$ **[M1]**

$15 + 8 = 23$ **[A1]**

(b) Substitutes values: $\frac{(-4)^2 - (2)^3}{2(3)}$ **[M1]**

Evaluates numerator: $16 - 8 = 8$

$\frac{8}{6} = \frac{4}{3}$ or $1\frac{1}{3}$ **[A1]**

[Total Marks: 4]

11 (a) (a) $40 + 25(3.5)$ or $40 + 87.5$ **[M1]**

127.50 **[A1]**

(b) $-12 + 9.8 \times 3$ **[M1]**

$-12 + 29.4$ **[M1]**

17.4 **[A1]**

[Total Marks: 5]

12 (a) $40 + 25h$ **[B1]**

(b) $(-2)^3 = -8$ and $(-3)^2 = 9$ **[M1]**

$\sqrt{\frac{1}{16}} = \frac{1}{4}$ **[B1]**

-68 **[A1]**

[Total Marks: 4]

13 Evaluates $p^{-2} = \left(\frac{1}{2}\right)^{-2} = 4$ **[B1]**

Evaluates $q^3 = (-2)^3 = -8$ **[B1]**

Evaluates $\sqrt[3]{r} = \sqrt[3]{-\frac{1}{8}} = -\frac{1}{2}$ **[B1]**

8 **[A1]**

[Total Marks: 4]

14 (a) $d = \sqrt{(-8-4)^2 + (2--3)^2}$ or $\sqrt{144+25}$ **[M1]**

13 **[A1]**

(b) $17 = \sqrt{(3-k)^2 + (-7-8)^2}$ **[M1]**

$289 = (3-k)^2 + 225$ **[M1]**

$(3-k)^2 = 64$ or $k^2 - 6k - 55 = 0$ **[M1]**

$k = -5$ and $k = 11$ **[A1]**

[Total Marks: 6]

15 (a) Forms three equations: $a+b+c=4$, $4a+2b+c=13$, $9a+3b+c=28$ **[M1]**

Eliminates c correctly to obtain two linear equations: $3a+b=9$ and $5a+b=15$ **[M1]**

$a = 3$ **[A1]**

$b = 0$ **[A1]**

$c = 1$ **[A1]**

(b) $[3(k+1)^2+1] - [3k^2+1] = 57$ or $6k+3=57$ **[M1]**

9 **[A1]**

[Total Marks: 7]

16 (a) States $A =$ Curved Surface Area + Base Area $= 2\pi rh + \pi r^2$ **[M1]**

Factorises πr out to obtain $\pi r(2h+r)$ **[A1]**

(b) (i) Substitutes $r=4$ and $h=6$: $A = \pi(4)(2(6)+4)$ **[M1]**

$A = 4\pi(12+4) = 64\pi$ **[A1]**

(ii) Rearranges volume formula: $h = \dfrac{V}{\pi r^2}$ **[M1]**

Substitutes h into area formula: $A = 2\pi r\left(\dfrac{V}{\pi r^2}\right) + \pi r^2$ **[M1]**

Simplifies first term: $\dfrac{2\pi rV}{\pi r^2} = \dfrac{2V}{r}$, resulting in $A = \dfrac{2V}{r} + \pi r^2$ **[A1]**

[Total Marks: 7]

17 (a) Substitute $x=-2$ and $y=4$: $P = \dfrac{2(-2)^2 - 3(4)}{-2-4}$ **[M1]**

$P = \dfrac{2(4)-12}{-6} = \dfrac{8-12}{-6}$ **[M1]**

$\frac{2}{3}$ (or equivalent) **[A1]**

(b) Multiply both sides by $(x-y)$: $P(x-y) = 2x^2 - 3y$ **[M1]**

Expand: $Px - Py = 2x^2 - 3y$ **[M1]**

Collect terms with y on one side: $3y - Py = 2x^2 - Px$ **[M1]**

Factorise and divide: $y(3-P) = 2x^2 - Px \implies y = \frac{2x^2 - Px}{3-P}$ (or equivalent) **[A1]**

[Total Marks: 7]

18 **(a)** **(i)** $\frac{6^2 - (-2)^2}{6(-2)}$ **[M1]**

$-\frac{8}{3}$ **[A1]**

(ii) $\frac{0.5^2 - 4^2}{0.5(4)}$ **[M1]**

$\frac{-15.75}{2}$ **[M1]**

$-\frac{63}{8}$ **[A1]**

(b) $\frac{x^2 - 2^2}{2x} = \frac{3}{2}$ **[M1]**

$2(x^2 - 4) = 6x$ (or $x^2 - 3x - 4 = 0$) **[M1]**

4 **[A1]**

[Total Marks: 8]

19 **(a)** $C + Rh$ or $C + hR$ **[B1]**

(b) M1 for $45 + 32.50 \times 3.5$ or $45 + 113.75$

158.75 **[A1]**

[Total Marks: 3]

20 Substitutes values correctly: $\frac{\sqrt{12.5 - 3.84}}{(-1.2)^3}$ **[M1]**

Evaluates numerator to 2.94... or denominator to -1.728 **[M1]**

-1.70 (accept $-1.703...$ rounded strictly to 3 sig figs) **[A1]**

[Total Marks: 3]

21 $\pi \times (4.5)^2 \times 8.2 + \frac{2}{3} \times \pi \times (4.5)^3$ **[M1]**

$521.6... + 190.8...$ or $712.5...$ **[M1]**

713 **[A1]**

[Total Marks: 3]

22 **(a)** $45 + 32h$ (or equivalent) **[B1]**

(b) 157 **[B1]**

(c) Correct substitution: $(-3)^3 - 4(-3)(2.5)$ or $-27 - (-30)$ **[M1]**

3 **[A1]**

[Total Marks: 4]

23 **(a)** $V = \frac{1}{3}\pi(6.4)^2(12.6)$ **[M1]**

540.5 **[A1]**

(b) $800 = \frac{1}{3}\pi r^2(14)$ **[M1]**

$r^2 = \frac{2400}{14\pi}$ or $54.567\ldots$ **[M1]**

7.39 **[A1]**

[Total Marks: 5]

24 **(a)** $85(7) + 0.24(450)$ **[M1]**

703 **[A1]**

(b) $1475.60 = 85(14) + 0.24m$ **[M1]**

$1475.60 - 1190 = 0.24m$ (or $285.60 = 0.24m$) **[M1]**

1190 **[A1]**

[Total Marks: 5]

25 **(a)** Substitute $h = 3.5$: $C = 45 + 35(3.5)$ **[M1]**

167.5 **[A1]**

(b) Substitute $C = 325$: $325 = 45 + 35h$ **[M1]**

$280 = 35h$ **[M1]**

8 **[A1]**

[Total Marks: 5]

26 **(a)** $A = \pi \times 5 \times \left(5 + \sqrt{5^2 + 12^2}\right)$ or 90π seen **[M1]**

283 (282.74...) **[A1]**

(b) $96\pi = \pi(6)\left(6 + \sqrt{6^2 + h^2}\right)$ **[M1]**

$16 = 6 + \sqrt{36 + h^2}$ or $\sqrt{36 + h^2} = 10$ **[M1]**

$36 + h^2 = 100$ or $h^2 = 64$ **[M1]**

8 **[A1]**

[Total Marks: 6]

27 **(a)** Substitutes $h = 4.2$ and $a = 6.5$ into formula: $V = \frac{1}{6}\pi(4.2)(3(6.5)^2 + 4.2^2)$ **[M1]**

$V = 317.53... = 317.5$ **[A1]**

(b) Multiplies by 6 and divides by πh: $\frac{6V}{\pi h} = 3a^2 + h^2$ **[M1]**

Subtracts h^2: $\frac{6V}{\pi h} - h^2 = 3a^2$ **[M1]**

$a^2 = \frac{2V}{\pi h} - \frac{h^2}{3}$ or $a^2 = \frac{1}{3}\left(\frac{6V}{\pi h} - h^2\right)$ **[A1]**

(c) $a = \sqrt{\frac{2(150)}{\pi(3)} - \frac{3^2}{3}} = \sqrt{31.83... - 3} = \sqrt{28.83...} = 5.37$ **[B1]**

[Total Marks: 6]

28 Numerator correctly evaluated: $-4.5 - (-3.2)^2 = -4.5 - 10.24 = -14.74$ **[B1]**

Denominator correctly evaluated: $1.8 - \sqrt{0.64} = 1.8 - 0.8 = 1$ **[B1]**

-14.74 **[B1]**

[Total Marks: 3]

29 **(a)** **(a)** $5.2(4.5) + 0.5(-9.8)(4.5)^2$ **[M1]**

-75.825 **[A1]**

(b) $-9.8 = 4.9t + \frac{1}{2}(-9.8)t^2$ **[M1]**

$4.9t^2 - 4.9t - 9.8 = 0$ or $t^2 - t - 2 = 0$ **[M1]**

$t = 2$ (rejects -1 as time must be positive) **[A1]**

[Total Marks: 5]

30 **(a)** M1 for correctly substituting the values into the formula: $\frac{1}{3}\pi(12.5)(4.2^2 + 4.2 \times 7.8 + 7.8^2)$

M1 for 111.24 or 463.5π or 1456.1...

1460 **[A1]**

(b) M1 for correctly substituting the values into the square root part: $\sqrt{(7.8 - 4.2)^2 + 12.5^2}$

M1 for $\sqrt{169.21}$ or 13.008...

736.9 **[A1]**

[Total Marks: 6]